

Hierarchical Prediction and Feedforward Correction of Time-Varying Thermal Line-of-Sight Bias for Coarse Acquisition in Satellite Optical Communications

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ABSTRACT

Thermal pointing bias can increase coarse acquisition search time in satellite optical communications before optical feedback becomes available. This study predicts the relative line-of-sight (LOS) angle between a star tracker reference and a laser communication terminal using the center temperature difference between the sun-facing and opposite panels and operational information, then applies the prediction as feedforward correction of the scan center. Thermoelastic analysis of a satellite with a box structure in low Earth orbit covered 21 cases spanning sun faces, internal dissipation, surface properties, and orbit conditions. The resulting hierarchical sun-face ΔT model has 16 coefficients. Panel temperature difference represents orbital variation, while biases based on sun face and dissipation state represent the DC terms for both axes. Nested leave-one-case-out evaluation excluded the test case from all coefficient estimates and gave a median RMSE of 4.9 μrad on the dominant axis over subsequent orbits, compared with a median raw LOS angle RMS of 615 μrad . In rectangular scan simulations for 14 cases under the baseline cold orbit and surface conditions, correction reduced the mean acquisition time from 14.6 to 0.10 s with thermal error only. With synthesized nonthermal errors, it reduced the mean from 19.2 to 5.45 s and increased the success rate from 98% to 100%. A preliminary study of two cases further showed that applying the previous orbit's observed residual to the next through a Fourier update reduced the periodic residual after feedforward correction.

Keywords: optical communication, pointing, acquisition, and tracking, thermal deformation, line-of-sight angle bias, feedforward correction

1. INTRODUCTION

Satellite optical communication provides high antenna gain and wide communication bandwidth, but its small beam divergence makes link establishment strongly dependent on pointing, acquisition, and tracking (PAT) performance¹⁾. During coarse acquisition, before stable optical feedback from the counterpart terminal is available, the spacecraft must scan an uncertainty region that includes orbit prediction error, attitude determination and control error, alignment residuals, terminal mounting error, and structural deformation²⁾. Because the number of search points and acquisition time increase

approximately with the uncertainty area, correcting predictable components before coarse acquisition can directly reduce link establishment time.

Initial pointing error sources differ substantially. With TLE orbit information, orbit prediction error can reach several hundred microradians in the link transverse plane¹⁾. Attitude determination and control error and calibrated alignment residuals can be limited to several tens of microradians in a system based on a star tracker²⁾. By contrast, thermal pointing shifts can be significant for coarse acquisition, depending on the spacecraft bus, optical design, and operating condition^{3–5)}. The analyses below likewise produce approximately 150 μrad to more than 1 mrad of thermal shift in LOS angle, depending on the sun face and internal dissipation, making it potentially one of the largest initial pointing error components.

This study focuses on the thermally induced component because it can be comparable to or larger than other initial error sources. That component is governed by known operating conditions, including eclipse cycles, the sun face, surface optical properties, and internal dissipation, so it is not a wholly unknown disturbance⁴⁾. The method below therefore predicts it from operating information rather than treating it as an unstructured search error. The resulting temperature field deforms the spacecraft and changes the relative attitude between the star tracker (STT), which defines the attitude reference, and the laser communication terminal (LCT), which defines the communication optical axis. Before optical feedback is available, this relative change appears as an offset of the coarse acquisition scan center. The model structure is not restricted to a particular orbit class, but low Earth orbit (LEO) is selected for evaluation because its short period and frequent illumination changes create a demanding thermal environment. Under such unstable illumination, periodic time series fitting has been reported to be difficult for LEO remote sensing⁶⁾. Demonstrating predictability in this demanding environment also supports the feasibility of applying the approach to more regular thermal environments.

Conventional PAT design absorbs initial pointing uncertainty through the scan range and beacon divergence^{1, 3)}. The corresponding search burden remains, however, and both thermal deformation and orbit prediction error can contain frequencies near the orbital period of approximately 100 min. Separating the thermal contribution from residuals observed after acquisition by frequency alone is therefore difficult. This motivates feedforward correction based on temperature and operating state before coarse acquisition. The central question is how much of the initial pointing error can be removed using information available before acquisition, rather than by retrospective residual separation.

Optical LOS angle correction using temperature has precedents. For the JUICE/JANUS camera, structural-thermal-optical-performance (STOP) analysis and ground testing established a relationship between a representative structural temperature difference and LOS angle variation^{7, 8)}. Thermal LOS angle correction based on orbital phase, thermal state, or observation geometry has also been reported for Earth observation satellites^{6, 9)}. Turella et al. further showed that overall optomechanical deformation, rather than individual optical elements, dominated LOS angle variation⁷⁾. It remains unclear whether spacecraft bus thermal deformation between a separated attitude reference and communication axis can be predicted across orbit, attitude, dissipation, and surface conditions and how such prediction changes coarse acquisition performance.

Accordingly, this paper asks whether (1) the thermally induced STT–LCT LOS angle can be predicted from a small set of temperature and operational variables, (2) model coefficients can be shared across conditions, and (3) scan center correction

based on that prediction improves acquisition performance. The study combines thermoelastic analyses of 21 conditions, a hierarchical sun-face ΔT model separating orbital variation from DC differences across conditions, nested leave-one-case-out validation, and PAT acquisition evaluation. The claims are limited to performance across conditions for the same box structure, STT/LCT placement, and LOS angle definition; universality of the numerical coefficients across other structures or flight hardware is not claimed.

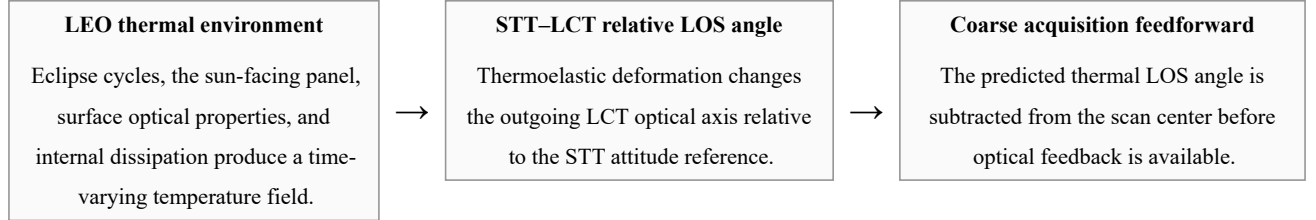


Figure 1. Problem addressed in this study. The thermally induced STT-LCT relative LOS angle is treated as a predictable component of the initial pointing error and is applied as feedforward correction of the scan center before coarse acquisition.

Nonthermal errors remain in the residual.

2. RELATED WORK AND POSITIONING OF THIS STUDY

Optical communication PAT studies have addressed pointing error budgets and scan design¹⁾, CubeSat demonstrations in orbit²⁾, feedforward compensation of attitude and mounting uncertainties¹⁰⁾, and beam pointing calibration after acquisition¹¹⁾. At the spacecraft bus level, Zhang et al. analyzed thermal pointing error between the STT and laser terminal references by finite element analysis⁵⁾, while Shi et al. reduced acquisition time by optimizing structural thermal stability and equipment placement³⁾. Those design approaches complement prediction of the residual thermal LOS angle from thermal state before acquisition.

LOS angle correction based on temperature has also been studied for Earth observation satellites and optical systems for deep space. Hu et al. used a diurnal model for a GEO remote sensor⁹⁾, whereas Li et al. used a neural network with observation geometry as its input under unstable LEO illumination⁶⁾. For JANUS, a wall-to-wall temperature difference was related to LOS angle through STOP analysis⁷⁾ and the coefficient was validated by ground testing⁸⁾. This paper does not claim novelty for a first-order relationship between temperature and LOS angle itself. Its novelty lies in extending that concept to the relative LOS angle between a separated STT and LCT on a spacecraft bus, hierarchically predicting DC terms that depend on the operating condition, and connecting the prediction to coarse acquisition performance.

Table 1. Representative prior studies and the position of this work. Optical communication studies do not predict thermal deformation error, whereas studies of Earth observation and optical instruments do not connect thermal correction to acquisition performance.

Study	System/domain	Treatment of thermal deformation error	Acquisition evaluation
Riesing et al. ²⁾	CubeSat optical PAT	Not modeled; absorbed by the attitude system	Yes; in orbit
Rüddenklau et al. ¹⁰⁾	Optical terminal	Not modeled; feedforward of attitude and mounting errors	Yes
Shi et al. ³⁾	Optical communication structural design	Reduced through structural design	Yes
Zhang et al. ⁵⁾	Inter-satellite optical pointing	Assessment by finite element analysis and design guidance	Yes; impact on link establishment
Hu et al. ⁹⁾	GEO Earth observation	Compensated by a periodic model	No
Li et al. ⁶⁾	LEO Earth observation	Learned and corrected by a neural network	No
Turella et al. ^{7, 8)}	Camera for deep space	Proportional model based on temperature difference	No
This study	Optical communication spacecraft bus evaluated in LEO	Prediction and correction by a hierarchical sun-face ΔT model	Yes

As summarized in Table 1, this work lies at the intersection of thermal LOS angle prediction from state and optical communication acquisition evaluation. JANUS addressed the change from a calibrated state within one optical head, whereas this study addresses the STT–LCT relationship across a spacecraft bus, 21 operating conditions, and DC components that are not absorbed by calibration. The present results remain numerical; validation of the coefficients on the ground and in orbit is future work.

3. PROBLEM DEFINITION AND LOS ANGLE CONVENTION

For far-field optical communication, the angular error relevant to coarse acquisition is the deviation of the outgoing LCT optical axis in inertial space. When the spacecraft attitude reference is provided by an STT, the STT’s own thermal rotation is absorbed into the rotation of the reference frame. The thermally induced communication axis error should therefore be defined as the rotation of the outgoing LCT optical axis relative to the STT attitude reference. Let $\theta_{\text{STT}} = (\theta_{\text{STT},x}, \theta_{\text{STT},y})$ and $\theta_{\text{LCT}} = (\theta_{\text{LCT},x}, \theta_{\text{LCT},y})$ denote the thermally induced rotations of the STT and LCT about two axes orthogonal to the nominal boresight. The relative rotation is

$$\theta_{\text{th}} = \theta_{\text{LCT}} - \theta_{\text{STT}} \quad (1)$$

and is hereafter called the far-field relative LOS angle and used as the primary output.

The centerline tilt obtained from the relative translation between the STT and LCT reference points is not added directly to the outgoing axis error because, for a far-field link, the angular contribution of translation is divided by the target range and is negligible. The far-field pointing error is instead governed by rotation of the local LCT optical axis relative to the STT rotation. Centerline tilt remains useful as a diagnostic measure of structural deformation modes and relative STT/LCT motion and is therefore reported as a secondary quantity.

Let θ_{nom} be the nominal pointing command and $\hat{\theta}_{\text{th}}$ the predicted relative LOS angle. The coarse acquisition scan center command is applied componentwise as

$$\theta_{\text{scan}} = \theta_{\text{nom}} - \hat{\theta}_{\text{th}}. \quad (2)$$

The residual after correction is the sum of the nonthermal error and thermal prediction error. The nonthermal term includes orbit prediction error, attitude determination and control error, alignment residuals, counterpart uncertainty, and unmodeled drift. The proposed method does not eliminate all these terms; it first removes the component predictable from thermoelastic analysis and temperature information to reduce the residual region searched during coarse acquisition.

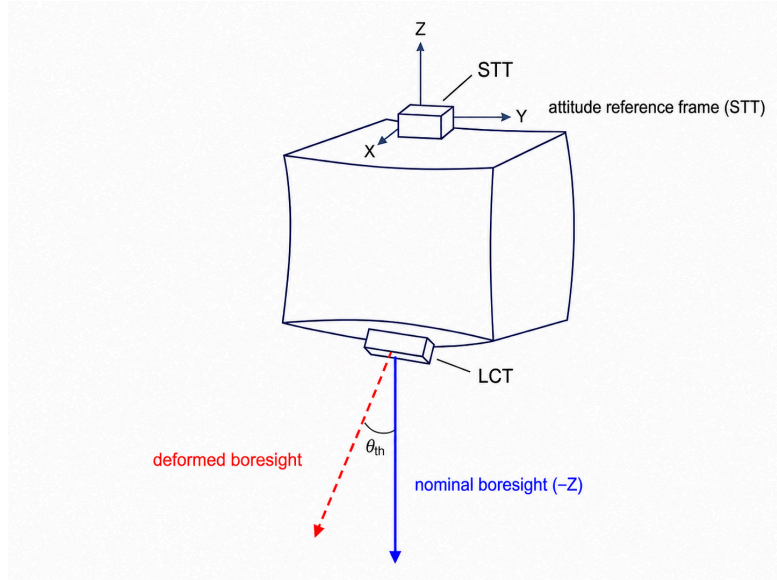


Figure 2. LOS angle convention. The outgoing LCT optical axis rotation relative to the STT attitude frame is used as the thermally induced LOS angle error corresponding directly to the far-field scan center error.

4. SPACECRAFT MODEL AND THERMOELASTIC ANALYSIS

4.1. Spacecraft model

The analysis uses a bus with a box structure representative of a small satellite (Figure 3 and Table 2). The STT and LCT, providing the attitude reference and communication, respectively, are mounted on the PZ and MZ panels, and PROP and

PCDU are included as internally dissipating units. Because the objective is evaluation across conditions rather than design review of a specific spacecraft, the basic structure, equipment placement, and LOS angle convention are fixed across all cases.

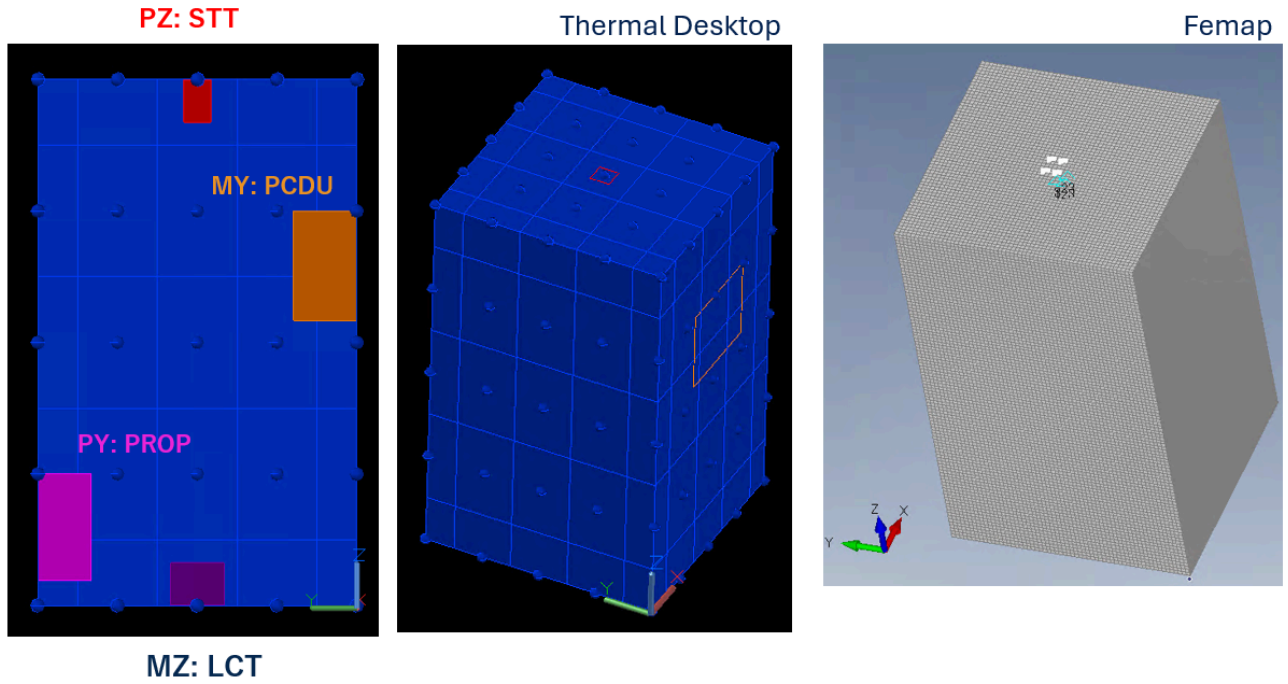


Figure 3. Spacecraft model with a box structure. Left: structural mesh in Femap. Right: Thermal Desktop thermal model showing an example temperature distribution in orbit. The STT is mounted on PZ, the LCT on MZ, and the boresight is approximately along the $-Z$ direction.

Table 2. Configuration of the spacecraft model with a box structure.

Item	Setting
Structure	Bus with a box structure, $0.6 \text{ m} \times 0.6 \text{ m} \times 1.0 \text{ m}$
Panel thickness	10 mm
Panel material	A5052 aluminum: Young's modulus 70.3 GPa, Poisson's ratio 0.33, coefficient of thermal expansion $2.38 \times 10^{-5} / \text{K}$, and reference temperature 24°C
Attitude reference (STT)	Mounted on PZ; 1.5 W dissipation, always ON
Communication terminal (LCT)	Mounted on MZ; 10 W dissipation, always ON; boresight approximately along $-Z$
Internal units	PROP: 25 W on PY; PCDU: 10 W on MY. ON/OFF states vary by case

4.2. Analysis procedure

The thermoelastic analysis consists of thermal analysis, structural response analysis, and LOS angle post-processing. Thermal Desktop first computes a periodic-steady-state temperature field for each case, which is mapped to the structural

model through TD Mapper Nastran temperature cards. Femap/Nastran then computes the six components of nodal displacement and rotation due to thermal deformation. The relative LOS angle time series is extracted from the rotations of the representative STT and LCT nodes at their centers. These nodal rotations are used as a proxy for equipment attitude; a least-squares rigid-body fit over the mounting-interface nodes was not applied and remains a future accuracy check. All conditions are managed by a case matrix and a common case identifier. The primary settings are listed in Table 3.

Table 3. Primary Thermal Desktop/Femap thermoelastic analysis settings. A neutral sun face is used in the baseline to compare sun face dependence without conflating coating differences.

Item	Setting
Orbit	Baseline: COLD case at LTAN06 and 800 km altitude, representing a cold mid-December case with $\beta = -58.3^\circ$ and eclipses. Comparisons: HOT, continuously illuminated; and an LTAN18 orbit at 693 km, similar to Sentinel-1
Duration and time step	18,157 s, approximately three 6,050 s orbits; 60.5 s time step
Thermal environment	COLD: solar irradiance 1309 W/m ² , albedo 0.2, and Earth infrared 189 W/m ² ; HOT/LTAN18: 1414 W/m ² , 0.4, and 261 W/m ²
Surface optical properties	Baseline: neutral sun face with $\alpha = \varepsilon = 0.5$, Black MY panel, and Alodine 1000 on the other panels; comparisons: Black sun face and Alodine on all surfaces
Initial/output thermal condition	Initial temperature 20 °C. Three orbits are output after Thermal Desktop periodic stabilization; initial transients are excluded from the evaluation
Temperature mapping	A temperature map is exported by TD Mapper and imported into Femap as Nastran temperature cards
Structural solution and constraints	Shell elements with a 3–2–1 minimal constraint at reference points near the STT mounting location
LOS angle output	Relative LOS angle components (x, y) from rotations of the representative STT and LCT nodes at their centers

In the deformation budget of a representative case, the mean centerline tilt between the STT and LCT reference points was approximately 206 μrad , the STT rotation contribution approximately 24 μrad , the LCT rotation contribution approximately 475 μrad , and the relative LOS angle approximately 460 μrad . Thus, the relative rotation of several hundred microradians between the STT and LCT is the dominant contribution to the far-field PAT LOS angle.

5. CASE MATRIX AND THERMAL LOS ANGLE CHARACTERISTICS

Twenty-one cases span sun faces, internal dissipation, surface optical properties, and orbit conditions (Table 4). The evaluated identifiers are 04–06 and 08–25. Cases 01–03 and 07 are MZ sun-face or setup analyses and are omitted because PZ and MZ, on which the STT and LCT are mounted, are not illuminated in the attitude family considered. The sun face

is selected from MX, MY, PX, and PY. The minimum dissipation configuration includes only the continuously operating STT and LCT; PROP and PCDU are then added separately or together, and one case uses half PROP power, 12.5 W. The PROP-only, PCDU-only, and no-additional-dissipation combinations for the four sun faces are split across Cases 13–21, Case 06, and Cases 23–24.

Table 4. Validation matrix of 21 cases, numbered 04–06 and 08–25. Unless noted otherwise, the COLD orbit and baseline surfaces are used. Additional dissipation excludes the continuously operating STT and LCT. Cases 13–21 cover nine of the twelve PROP-only / PCDU-only / no-additional combinations; the remaining three are Case 06 and Cases 23–24. Cases 01–03 and 07 are MZ sun-face or setup runs and are excluded because PZ and MZ are not illuminated in this attitude family.

Case	Purpose	Conditions
04–06, 08–09	Sun face and baseline dissipation	MX/MY/PX/PY; all units dissipating or STT/LCT only
10	Orbital thermal environment	MY, all units dissipating, HOT with continuous illumination
11–12	Surface optical properties	MY; Black sun face or Alodine on all surfaces
13–21	Dissipation mode	Partial 4-face set of PROP only, PCDU only, or no additional dissipation
22	Continuous dissipation level	MY; PROP at half power, 12.5 W, plus PCDU
23–24	MX dissipation mode	MX; PROP only or PCDU only
25	Orbit condition	MY, all units dissipating, LTAN18 and 693 km

Figure 4 shows the panel temperatures and relative LOS angle for representative Case 04, with MY facing the Sun and all units dissipating. Both temperature and thermal LOS angle vary at the orbital period, and the MY panel temperature follows the y component of the LOS angle closely. This observation motivates the panel temperature difference ΔT used below.

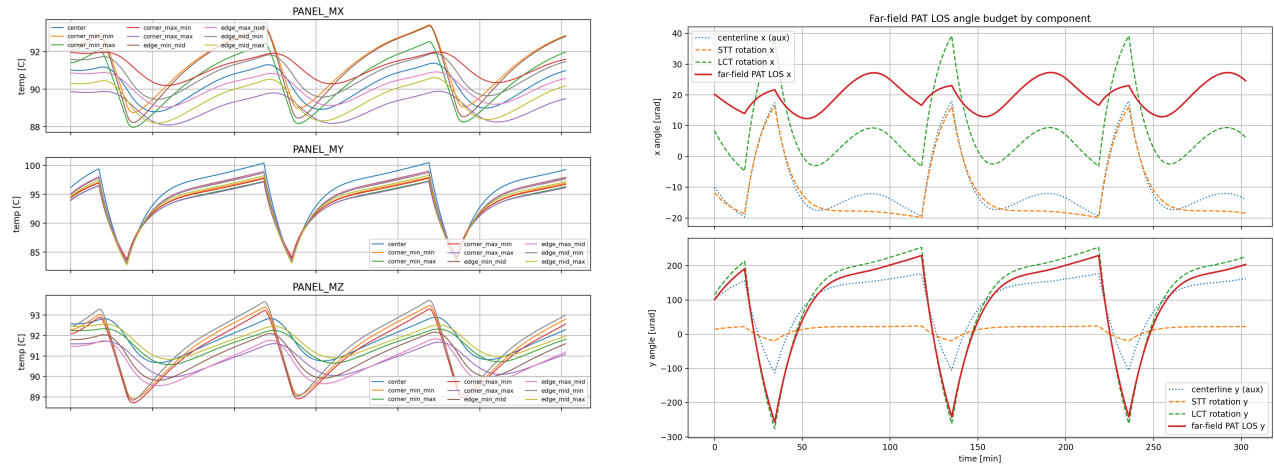


Figure 4. Representative Case 04, with MY facing the Sun and all units dissipating. Left: representative panel temperatures. Right: relative LOS angle components. The temperature field and LOS angle vary at the same orbital period, and the MY panel temperature follows the y component of the LOS angle.

Three observations summarize the 21 cases. Hereafter, the axis containing most of the orbital variation of the relative LOS angle is termed the dominant axis. First, the raw dominant axis RMS depends strongly on the sun face: approximately 150–265 μrad for MY, 600–670 μrad for PX, 670–730 μrad for MX, and 1180–1280 μrad for PY. Second, the dominant axis switches systematically: it is y for MY/PY and x for PX/MX. Third, surface optical properties affect the variation amplitude and residual floor, with a higher floor for a Black coating, whereas the internal dissipation layout mainly changes the mean bias across cases.

These observations impose two requirements on a reduced-order model. The sun face must be explicit because it changes the dominant axis and sign, and continuous orbital variation must be separated from the offset across cases associated with dissipation mode. The hierarchical sun-face ΔT model below is the smallest model constructed to satisfy both requirements.

6. HIERARCHICAL SUN-FACE ΔT MODEL

Running a high-fidelity Thermal Desktop/Femap analysis onboard before every coarse acquisition attempt is impractical. The Thermal Desktop/Femap thermal LOS angle is therefore treated as truth and reduced to a model computable from a small number of temperature and operational inputs. The purpose is not merely to minimize regression error, but to obtain physically interpretable inputs suitable for onboard use.

The most stable decomposition found in the study is as follows. Orbital variation is represented by the difference $\Delta T(t)$ between the center temperatures of the sun-facing and opposite panels. Local internal dissipation effects, including PROP and PCDU, are treated as constant case biases rather than additional orbital temperature features. Extensions using mounting point temperatures and local temperature differences were also examined, but their coefficients were unstable because of collinearity with $\Delta T(t)$ and low signal-to-noise ratio. The adopted hierarchy therefore assigns orbital variation to ΔT and the remaining dissipation effect to a case bias.

For each case, the panel temperature difference is

$$\Delta T(t) = T_{\text{sunface}(t)} - T_{\text{opposite}(t)}. \quad (3)$$

The relative LOS angle θ_{th} is the vector with two components defined by Equation 1. The component containing most orbital variation is denoted θ_{dom} ; the corresponding component of $\hat{\theta}_{\text{th}}$ is denoted $\hat{\theta}_{\text{dom}}$. Its thermal LOS angle is approximated within a case as

$$\theta_{\text{dom}(t)} \approx b_{\text{case}} + a(\text{sun})\Delta T(t), \quad (4)$$

where $a(\text{sun})$ is the sensitivity for each sun face in $\mu\text{rad/K}$ and b_{case} is the case DC bias in microradians. The bias is required because the LOS angle need not be zero at $\Delta T = 0$ and because mounting and local dissipation leave residual terms.

The non-dominant axis has only several microradians RMS variation about its case mean and is therefore modeled without a time-varying term; only its constant case DC bias b_{nd} is corrected. That bias nevertheless depends systematically on the sun face and is non-negligible: approximately $-600 \mu\text{rad}$ for MX/PX and $+20 \mu\text{rad}$ for MY/PY in this model. Setting it to

zero severely degrades coarse acquisition performance, so b_{nd} is predicted by the same Level 2 framework as the dominant axis bias.

The dominant and non-dominant axis biases, b_{case} and b_{nd} , are modeled across cases using sun face dummy variables and dissipation flags:

$$b_{case} \approx b_0(\text{sun}) + c_{prop}I_{prop} + c_{pcdu}I_{pcdu}. \quad (5)$$

Here I_{prop} and I_{pcdu} equal one when the corresponding unit is ON and zero otherwise. Additional dissipating units can be incorporated by adding analogous terms, so the parameter count increases linearly with the number of units. The form in Equation 5 is fitted independently for the two axes. Because PROP and PCDU are mounted on the $\pm Y$ panels, their effect appears primarily in the y component of the LOS angle. The flag terms are therefore enabled for the dominant axis bias in MY/PY cases, where y is dominant, and for the non-dominant axis bias in MX/PX cases, where y is non-dominant. For prediction, the median of the independently fitted a_{emp} values for each sun face is used as the shared sensitivity $a_{shared(\text{sun})}$:

$$\hat{\theta}_{dom(t)} = b_{pred(\text{sun}, I_{prop}, I_{pcdu})} + a_{shared(\text{sun})}\Delta T(t). \quad (6)$$

The fixed model contains 16 scalar coefficients: four a coefficients, eight b_0 coefficients for the two axes, and four dissipation flag coefficients for the two axes.

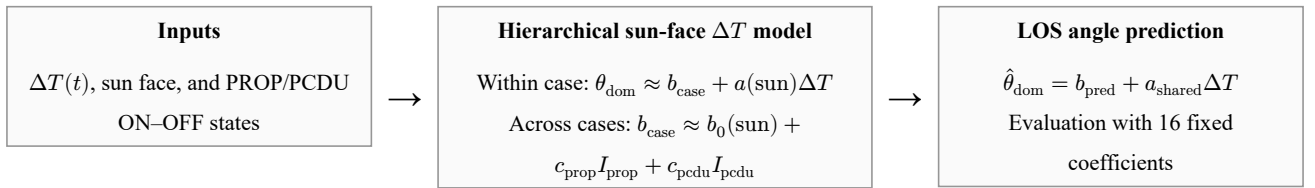


Figure 5. Structure of the hierarchical sun-face ΔT model. The panel temperature difference $\Delta T(t)$ represents orbital variation, while sun face dummy variables and internal dissipation flags predict the DC bias across cases not captured by ΔT .

The coefficients are identified in two stages. First, Equation 4 is fitted to the first orbit of each case to estimate empirical a_{emp} and b_{emp} values; for the non-dominant axis, the training orbit mean is used as the empirical b_{nd} . Second, Equation 5 is fitted by ordinary least squares across cases. Generalization is evaluated by nested leave-one-case-out (nested LOO): the test case is removed from the Level 2 bias fit, from the calculation of a_{shared} , and from the non-dominant axis b_{nd} fit. Thus, no coefficient specific to the evaluated case is known in advance.

Across the 21 cases, the shared sensitivities for MX, MY, PX, and PY were +30.6, +28.6, −28.1, and −28.7 $\mu\text{rad/K}$, respectively (Table 5). Their magnitudes cluster at approximately 28–31 $\mu\text{rad/K}$, and their signs follow the sun face and dominant axis direction. The coefficients were not forced to agree; the independently estimated values clustered by sun face, supporting their use as shared coefficients for this spacecraft model, LOS angle definition, and case set.

Table 5. Identified coefficients of the hierarchical sun-face ΔT model. The dissipation flag coefficients were $c_{\text{prop}} = -22.9 \mu\text{rad}$ and $c_{\text{pctu}} = -10.2 \mu\text{rad}$ for the dominant axis and $c_{\text{prop}} = -131 \mu\text{rad}$ and $c_{\text{pctu}} = +22.1 \mu\text{rad}$ for the non-dominant axis. The flag terms apply only to MY/PY cases on the dominant axis and MX/PX cases on the non-dominant axis.

Coefficient	MX	MY	PX	PY
Shared sensitivity a_{shared} [$\mu\text{rad/K}$]	+30.6	+28.6	-28.1	-28.7
Baseline bias b_0 , dominant axis [μrad]	+15.7	+2.8	-12.0	-24.0
Baseline bias b_0 , non-dominant axis [μrad]	-594	+16.6	-596	+25.1

The dissipation coefficients represent the DC residual after the panel temperature difference contribution has been removed; they do not directly indicate the physical deformation direction caused by heating. Level 2 also reproduced the non-dominant axis bias across sun faces, with a nested LOO RMSE of $3.1 \mu\text{rad}$ and a maximum error of $5.3 \mu\text{rad}$.

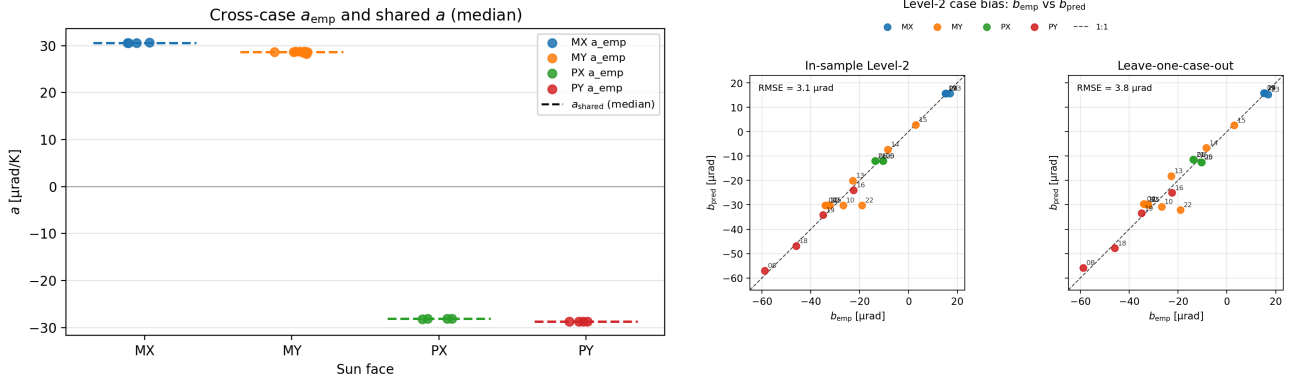


Figure 6. Behavior of the hierarchical sun-face ΔT model across conditions. Left: independently estimated a_{emp} values cluster by sun face and support a shared sensitivity. Right: empirical b_{emp} versus Level 2 b_{pred} on the dominant axis. The dominant-axis bias RMSE over the 21 cases is $3.1 \mu\text{rad}$ in-sample and $3.8 \mu\text{rad}$ under leave-one-case-out. The nested LOO RMSE of the non-dominant-axis bias is $3.1 \mu\text{rad}$.

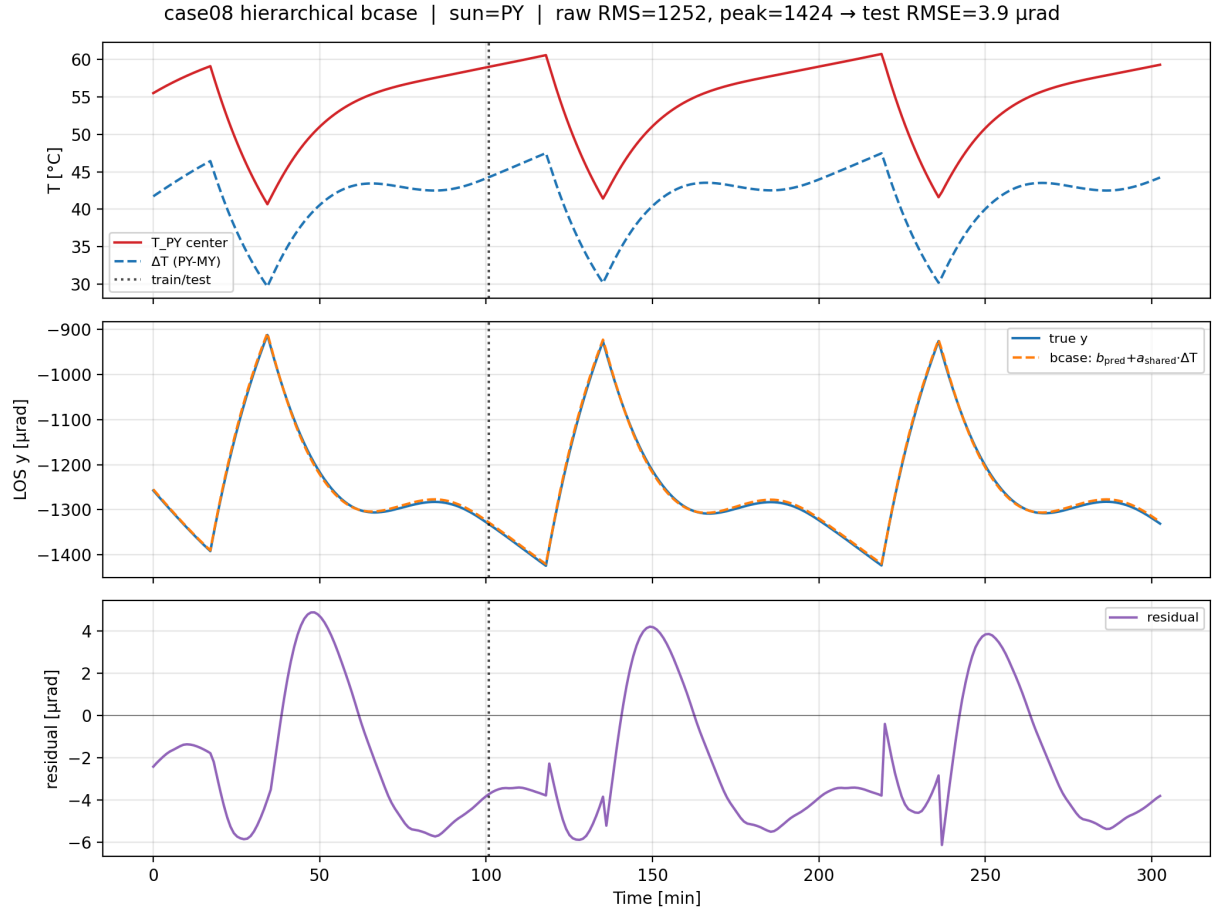


Figure 7. Example prediction for Case 08, with PY facing the Sun and all units dissipating. The raw dominant-axis RMS is 1250 μrad and the test RMSE is 3.9 μrad . The model follows the orbital thermal LOS angle variation using $b_{\text{pred}} + a_{\text{shared}} \Delta T(t)$.

For the standard COLD orbit and baseline surfaces, the dominant axis test RMSE was generally 3–7 μrad . In Case 08, with PY facing the Sun and all units dissipating, the raw RMS was 1250 μrad and the RMSE after model correction 3.9 μrad . For MX, a raw RMS of approximately 670–730 μrad was reduced to approximately 3 μrad ; for PX, approximately 600–670 μrad was reduced to 5–6 μrad . The MY dissipation series had a smaller raw RMS of approximately 150–260 μrad and an RMSE after model correction of 6–7 μrad , close to the Level 1 time-varying residual floor. Across all 21 cases, the nested LOO dominant axis test RMSE had a median of approximately 4.9 μrad and a mean of approximately 5.5 μrad , one to two orders of magnitude below the median raw RMS of approximately 615 μrad . The LTAN18/693 km case similar to Sentinel-1 also had a small test RMSE of approximately 1.2 μrad , indicating that the model structure remained effective under the different orbit condition.

The limitations are also clear. A Black coating increased the residual variation to approximately 13 μrad , and the HOT orbit left a DC offset of several microradians. In the case with PROP at half power, the ON/OFF flag assigned the same input as PROP at full power and could not represent continuous dissipation, increasing the nested LOO residual to approximately

16 μrad . The main result is therefore the reduction by one to two orders of magnitude obtained with 16 fixed coefficients, centered on the standard COLD conditions. Explicit treatment of coating, orbit, and continuous dissipation remains future work.

7. PAT COARSE ACQUISITION EVALUATION

To evaluate communication system effects, the thermal LOS angle prediction is connected to a coarse acquisition simulator using a rectangular spiral. The simulator takes the Thermal Desktop/Femap thermal LOS angle as truth and computes acquisition success and time for each scan center correction method. Acquisition occurs when the true target direction lies within the detection radius of a scan point. Acquisition time is therefore a practical proxy for the residual initial pointing uncertainty after correction.

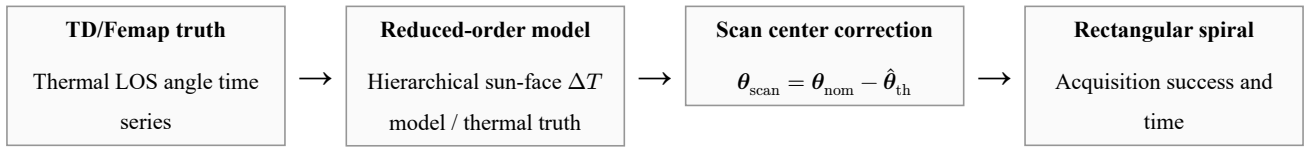


Figure 8. PAT evaluation flow. The Thermal Desktop/Femap LOS angle is used as thermal truth, connected to feedforward scan center correction by the reduced-order model, and evaluated through acquisition time in a rectangular spiral scan.

7.1. Scan conditions

The scan conditions in Table 6 represent coarse acquisition with a beacon instantaneous field of view rather than the narrow communication beam. Figure 9 shows a local excerpt of the rectangular spiral. A detection radius of 150 μrad is half the width of the approximately 0.3 mrad beacon field, and a 120 μrad step provides 60% overlap³⁾. Half the grid diagonal, $\frac{120}{\sqrt{2}} \approx 85 \mu\text{rad}$, is smaller than the detection radius, so the scan region has no coverage holes. A range of $\pm 1600 \mu\text{rad}$ allows margin for the largest thermal LOS angle, approximately 1.2–1.3 mrad for PY, combined with nonthermal error. These times are proxies: a 0.2 s dwell as in Shi et al. would double all times but preserve their relative ordering.

Table 6. Coarse acquisition scan conditions. A rectangular spiral is assumed; slew and settling between scan points and stochastic received power variation are neglected.

Item	Value	Basis
Scan range	$\pm 1600 \mu\text{rad}$	Covers the approximately 1.3 mrad maximum thermal LOS angle plus nonthermal error
Scan step	120 μrad	60% overlap for an approximately 0.3 mrad beacon field
Detection radius	150 μrad	Beacon coarse field of view ³⁾
Dwell time	0.1 s/point	Representative beacon integration time
Number of points	729, or 27×27	Maximum full scan time of 72.9 s

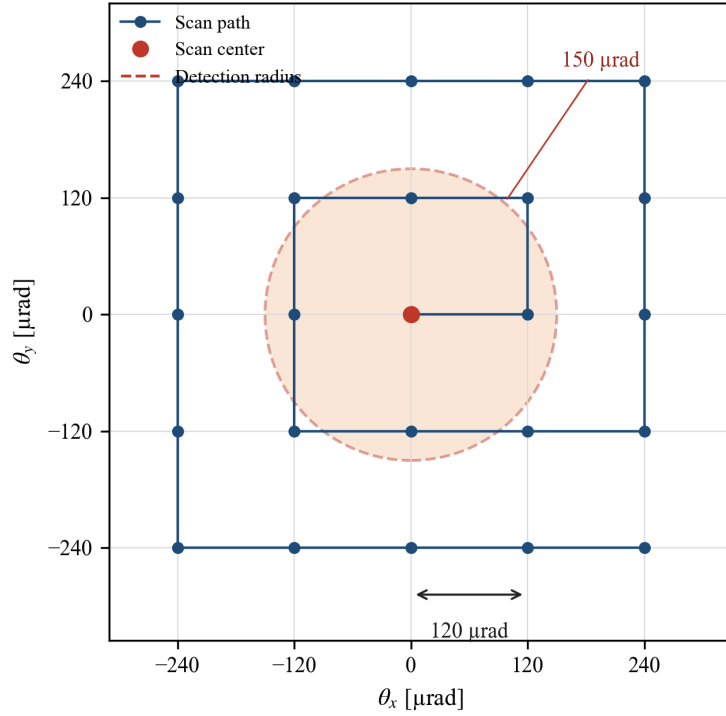


Figure 9. Local excerpt of the rectangular spiral used for coarse acquisition. Adjacent points are $120\ \mu\text{rad}$ apart, and the dashed circle is the $150\ \mu\text{rad}$ detection radius. The evaluation uses a 27×27 grid over $\pm 1600\ \mu\text{rad}$.

7.2. Nonthermal error model

A synthesized nonthermal error represents a more operationally representative condition. The latest Sentinel-1 TLE is propagated with SGP4; its position error relative to a precise orbit ephemeris (POEORB) is projected onto the link transverse plane, converted to angular error in the STT/body x–y frame, and mapped periodically onto the evaluation interval. The model also includes a constant alignment residual with $1\sigma = 50\ \mu\text{rad}$, random attitude determination and control error with $1\sigma = 50\ \mu\text{rad}$, and low-frequency drift of $30\ \mu\text{rad}$ amplitude and $900\ \text{s}$ period. Rather than reproducing a specific spacecraft error budget, this synthesis represents coexisting errors on a small LEO satellite without GNSS. Both the thermal LOS angle and orbit prediction error vary at frequencies near the orbital period of approximately $100\ \text{min}$, so frequency separation after observation cannot isolate the thermal component. This motivates feedforward correction based on temperature and operational state.

7.3. Results

We compare no correction, correction by the hierarchical sun-face ΔT model in Equation 6, and thermal truth correction as an ideal upper bound. Nested LOO excludes the evaluated case from the shared sensitivity and both Level 2 axis bias estimates. We evaluate 14 COLD cases with baseline surfaces, Cases 4–6, 8–9, and 13–21. For each case, acquisition is scored at every output epoch of the three-orbit series after Thermal Desktop periodic stabilization, 301 samples at $60.5\ \text{s}$.

Reported times and success rates are means of the 14 case means. The synthesized nonthermal error uses one fixed seed per case and is not a Monte Carlo ensemble.

Table 7. Mean coarse acquisition performance over 14 COLD baseline-surface cases (Cases 4–6, 8–9, and 13–21). Each case mean uses 301 epochs over three orbits after periodic thermal stabilization. Nonthermal error is one seed per case, not a Monte Carlo ensemble. With thermal error only, correction by the hierarchical sun-face ΔT model requires one scan point, one dwell of 0.1 s, and matches the ideal thermal truth upper bound. With nonthermal error, it reduces the mean acquisition time from 19.2 to 5.45 s, approximately 72%, and increases the success rate from 98% to 100%.

Condition	No correction	Hierarchical sun-face ΔT model	Thermal truth
Thermal only: mean acquisition time	14.6 s	0.10 s	0.10 s
Thermal only: success rate	100%	100%	100%
With nonthermal error: mean acquisition time	19.2 s	5.45 s	—
With nonthermal error: success rate	98%	100%	—

With thermal error only, the hierarchical sun-face ΔT model yields a mean acquisition time of 0.10 s, matching thermal truth correction because acquisition occurs at the first scan point. The mean thermal residual across the two axes after correction is 9.3 μrad , well below the 150 μrad detection radius. The model removes the case DC term through b_{pred} and the orbital variation through $a\Delta T(t)$.

With nonthermal error, correction reduces the mean acquisition time from 19.2 to 5.45 s. The mean initial error after correction is 448 μrad and is almost entirely dominated by nonthermal error. The role of the method is therefore not to eliminate total pointing error, but to remove the large time-varying thermal bias and reduce the search burden to the nonthermal error level.

The improvement varies by sun face. It is small for MY, whose thermal LOS angle is 150–260 μrad , while the large orbit prediction error for PX/MX, whose thermal LOS angle is 0.6–0.9 mrad, determines the floor after correction. For PY, whose thermal LOS angle is approximately 1.2 mrad, the acquisition time and success rate improve from 37.3–39.5 s and 88–97% to 1.3–1.8 s and 100%, respectively. The benefit is thus largest under the most severe thermal deformation conditions.

7.4. Preliminary residual update

The primary feedforward (FF) method uses only temperature and operational information available before acquisition. This subsection is a numerical experiment: it asks whether a periodic residual remaining after FF can be carried to the next orbit if a residual time series is available after acquisition. The assumed observable is the total pointing residual on a receiver sensor after successful acquisition, not an isolated thermal component.

As a supplementary study, the residual on the dominant axis after acquisition, $r(t) = \theta_{\text{dom}(t)} + e_{\text{nth}(t)} - \hat{\theta}_{\text{dom}(t)}$, is represented by a Fourier series in orbital phase $\varphi = 2\pi t/T_{\text{orb}}$, and coefficients identified during orbit n are applied to orbit $n + 1$. This is not re-identification of the thermal model, but a preliminary operational correction of periodic error remaining after FF:

$$\hat{r}(\varphi) = c_0 + \sum_{k=1}^K [a_k \cos(k\varphi) + b_k \sin(k\varphi)]. \quad (7)$$

The $K = 2$ coefficients are estimated by ridge regression, with no update applied during the first orbit. Two representative cases are compared against direct Fourier fitting of total error, in which Fourier fitting is applied to the total error without FF (Table 8).

Table 8. Comparison of residual updates with nonthermal error, causal fitting, and $K = 2$. Direct Fourier fitting of total error reaches the floor from orbit 1 onward but is uncorrected in the first orbit; for the large thermal LOS angle of PY, its acquisition time in the first orbit exceeds the requirement of approximately 30 s. FF + residual Fourier preserves FF performance from the first orbit and reduces the floor to approximately 0.4 s from orbit 1 onward.

Method	All data	First orbit, orbit 0	Orbit 1 onward
Case 13 (MY)			
FF only	1.64 s	1.60 s	1.66 s
Direct Fourier fitting of total error	1.39 s	3.37 s	0.39 s
FF + residual Fourier	0.79 s	1.60 s	0.38 s
Case 16 (PY)			
FF only	1.55 s	1.47 s	1.59 s
Direct Fourier fitting of total error	13.2 s, 98.7% success	39.5 s	0.38 s
FF + residual Fourier	0.76 s	1.47 s	0.40 s

From orbit 1 onward, both Fourier approaches reach approximately 0.4 s. Direct Fourier fitting of total error, however, is uncorrected in the first orbit and requires 39.5 s for Case 16, whose thermal LOS angle is approximately 1 mrad. Combining FF with the residual update preserves FF performance in the first orbit and reduces the periodic floor in subsequent orbits. A separate update of only the constant residual component reduced the DC residual but left the AC floor at the orbital period, improving Case 13 from 1.66 to 1.35 s from orbit 1 onward, compared with 0.38 s for residual Fourier. This difference provides the quantitative basis for using a periodic model in the second layer. Nevertheless, this evaluation uses only two cases, with all samples at intervals of 60.5 s, residuals at failed acquisition points, and periodically mapped orbit error. It does not implement a specific receiver sensor or sparse sampling. Hardware realization of the residual observation, and performance with samples available only after successful acquisitions, remain future work.

Operationally, coarse acquisition for a particular counterpart and orbit is often effectively a first encounter. The FF prediction by the hierarchical sun-face ΔT model can be applied immediately from temperature and operating state without residual learning. After successful acquisitions have accumulated residual observations, recurrent components can be carried to later opportunities. This ordering avoids a burdensome first acquisition while permitting a lower recurrent residual floor.

8. DISCUSSION AND CONCLUSION

The temperature difference between the panel facing the Sun and the opposite panel represents structural bending during an orbit, while the bias that remains constant within each case represents DC differences among conditions caused by local dissipation and related effects. This separation represents the thermal LOS angle across multiple sun faces and dissipation modes with 16 coefficients. The contribution is not the first-order temperature model itself, but its application to the STT–LCT relative LOS angle at the spacecraft bus level, coefficient sharing across conditions, and connection to coarse acquisition time.

The shared sensitivity is an empirical coefficient specific to the present structure, placement, LOS angle definition, and case set, and must be re-identified for another configuration. In operation, ΔT is assumed to be obtained from temperature sensors at the centers of the panel facing the Sun and the opposite panel. With a sensitivity of approximately 30 $\mu\text{rad}/\text{K}$, an error of 0.1 K in the temperature difference corresponds to approximately 3 μrad . Sensor placement, delay, and quantization nevertheless require future error analysis. Orbit, attitude, and alignment errors also leave a floor of several hundred microradians after thermal correction, so the method is intended to reduce the search region in conjunction with treatment of those errors.

The evaluation is limited to numerical analysis of one box structure and includes no ground test. Residuals increase for some coating and orbit conditions, and both the nonthermal error and scan models are simplified. Slew and settling between scan points and probabilistic detection are neglected. The residual Fourier update remains a preliminary evaluation of two cases with dense sampling.

In conclusion, the median nested LOO RMSE on the dominant axis is 4.9 μrad , compared with a median raw LOS angle RMS of 615 μrad . Across 14 COLD baseline-surface cases, the mean acquisition time is reduced from 14.6 to 0.10 s with thermal error only and from 19.2 to 5.45 s with nonthermal error. Thus, under the evaluated conditions, feedforward correction reduces the predictable thermal component until nonthermal errors govern the residual search burden, contributing to faster satellite optical link establishment before optical feedback is available. Future work will incorporate coating, orbit, and continuous dissipation into Level 2, validate the coefficients by ground testing, and extend the residual update to sparse successful observations and all cases.

REFERENCES

- 1) Kaushal, H., Kaddoum, G. Optical Communication in Space: Challenges and Mitigation Techniques. *IEEE Communications Surveys & Tutorials*. 2017, vol. 19, no. 1, pp. 57–96.
- 2) Riesing, K. M., Schieler, C. M., Chang, J. S., Gilbert, N. A., Horvath, A. J., Reeve, B. S., Robinson, B. S., Wang, J. P., Agrawal, P. S., Goodloe, R. A. “On-orbit results of pointing, acquisition, and tracking for the TBIRD CubeSat mission.” *Proc. SPIE*. 2023.
- 3) Shi, Y., Chen, S., Yu, M., Wu, Y., Yu, J., Zhang, L. Thermal Deformation Stability Optimization Design and Experiment of the Satellite Bus to Control the Laser Communication Load's Acquisition Time. *Applied Sciences*. 2023, vol. 13, no. 9, p. 5502.
- 4) Badás, M., Piron, P., Bouwmeester, J., Loicq, J., Kuiper, H., Gill, E. Opto-thermo-mechanical phenomena in satellite free-space optical communications: survey and challenges. *Optical Engineering*. 2024, vol. 63, no. 4, p. 41206.
- 5) Zhang, P., Cheng, F., Guan, Z., He, W., Han, Y. Analysis and Study on the Impact of Rapid Link Establishment of Inter-Satellite Laser in External Heat Flux Environment. *Journal of Telemetry, Tracking and Command*. 2025, vol. 46, no. 2, pp. 20–33.
- 6) Li, Y., Chen, X., Liu, G., Rao, P. Correction Method for Thermal Deformation Line-of-Sight Errors of Low-Orbit Optical Payloads Under Unstable Illumination Conditions. *Remote Sensing*. 2025, vol. 17, no. 5, p. 762.
- 7) Turella, Andrea, Della Corte, Vincenzo, Paolinetti, Riccardo, Palumbo, Pasquale, Amoroso, Marilena, Castronuovo, Marco, Mugnuolo, Raffaele. “Structural-Thermal-Optical-Performance (STOP) analysis for the prediction of the line-of-sight stability of JANUS camera on board JUICE ESA mission.” *Proc. SPIE 11103, Optical Modeling and System Alignment*. 2019, p. 111030E.
- 8) Turella, Andrea, Della Corte, Vincenzo, Palumbo, Pasquale, Amoroso, Marilena, Mugnuolo, Raffaele, Noci, Giovanni Enrico. “JANUS Optical Head Line of Sight Temperature dependence Characterization and Validation by on ground test.” 2021 IEEE 8th International Workshop on Metrology for AeroSpace (MetroAeroSpace). 2021.
- 9) Hu, H., Tong, X., Qiu, C., Li, H., Fu, J., Shang, Y., Shi, J. On-Board Thermal Motion Compensation Method for Pointing Errors of the Remote Sensor Aboard a Three-Axis Stabilized Geostationary Satellite. *IEEE Geoscience and Remote Sensing Letters*. 2022, vol. 19, p. 6508405.
- 10) Rüddenklau, R., Barone, S., Garbagnati, E., Spier, S., Schitter, G. Feed-Forward Compensation of Body-Pointing Uncertainties for Laser Communication Terminals. *Journal of Guidance, Control, and Dynamics*. 2026, vol. 49, no. 6, pp. 1810–1819.
- 11) Cierny, O., Cahoy, K. L. On-orbit beam pointing calibration for nanosatellite laser communications. *Optical Engineering*. 2019, vol. 58, no. 4, p. 41605.